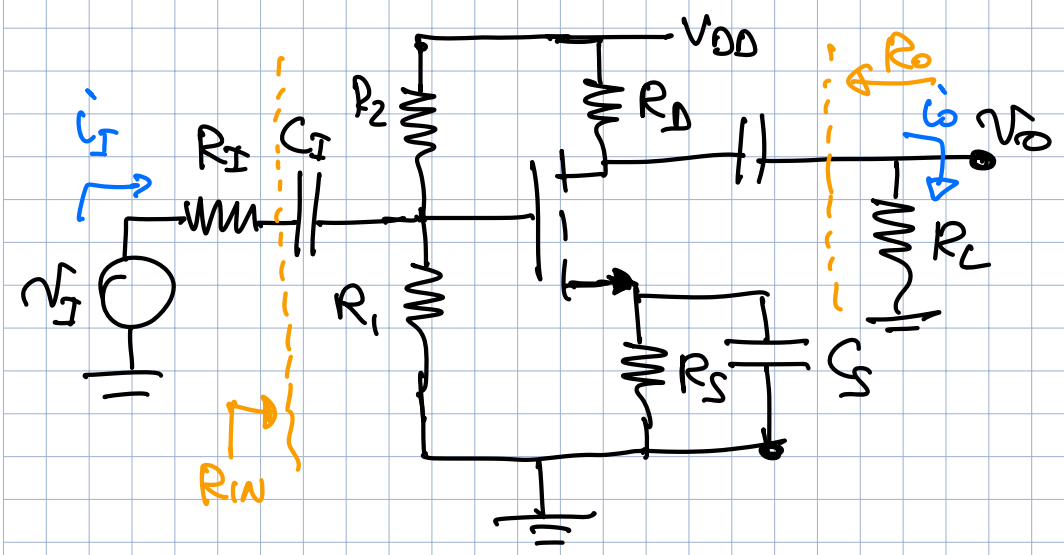
ESERCIZIO A1

AMPLIFICATORE A SOURCE COMUNE



$V_{DD} = 15V$

$V_{TN} = 1V$,

$K_n = 0{,}25\ mA/V^2$

$R_I = 1\ K\Omega$

$R_1 = 1\ M\Omega$

$R_2 = 2{,}7\ M\Omega$

$R_D = 82\ K\Omega$

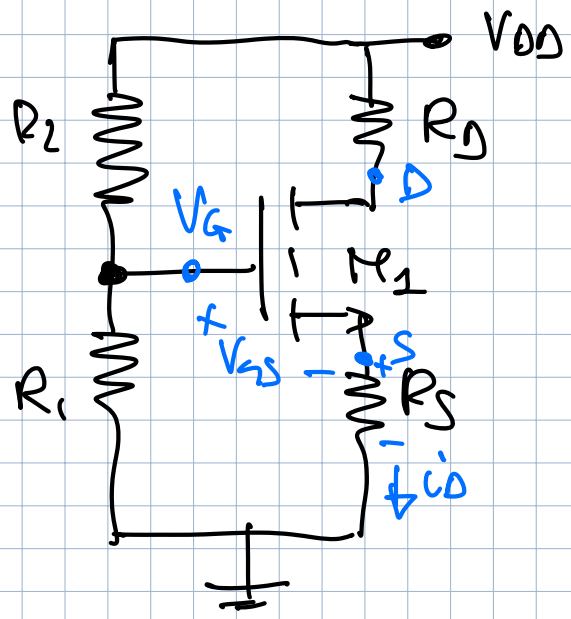
$R_S = 27\ K\Omega$

$R_L = 5\ K\Omega$

$\lambda = 0{,}05 V^{-1}$

1) TROVARE IL PUNTO DI LAVORO
   I_D, V_{DS}, V_{GS} ?
2) RESISTENZE DI INGRESSO e USCITA
3) GUADAGNO DI TENSIONE $A_V = \frac{v_O}{v_I}$
4) GUADAGNO DI CORRENTE $A_I = \frac{i_O}{i_I}$

ANALISI DC (CALCOLO PUNTO DI LAVORO)



$$V_G = V_{DD} \cdot \frac{R_1}{R_1 + R_2} = 4{,}05V$$

$V_G > V_{TN} \Rightarrow$ MOSFET "ON"

HP MOSFET SATURAZIONE

$$\begin{cases} I_D = \frac{K_n}{2}(V_{GS} - V_{TN})^2 \\ V_G = V_{GS} + R_S I_D \end{cases}$$

NON USO λ!

$$\begin{cases} I_D = \dfrac{V_G - V_{GS}}{R_S} \\ \dfrac{2V_G}{R_S K_n} - \dfrac{2V_{GS}}{R_S K_n} = V_{GS}^2 + V_{TN}^2 - 2V_{GS}V_{TN} \end{cases}$$

$$V_{GS}^2 + V_{GS}\left(\frac{2}{R_S K_n} - 2V_{TN}\right) + V_{TN}^2 - \frac{2V_G}{R_S K_n} = 0$$

$a = 1 \qquad b = -1{,}70 \qquad c = -0{,}2$

$$V_{GS} = \begin{cases} +1{,}81\,V & \text{OK} \\ -0{,}11\,V & \leftarrow \text{NON VALIDA } (< V_{TN}) \end{cases}$$

$I_D = 0{,}082\ mA = 82\ \mu A$

$V_{DS} = V_{DD} - (R_D + R_S) I_D = 6{,}06\ V$

$V_{DS} > V_{GS} - V_{TN}$?

$6{,}06 > 0{,}81\ V$ Sì: OK MOSFET SATURAZIONE

$M_1: (I_D = 82\ \mu A;\ V_{DS} = 6{,}06\ V)$

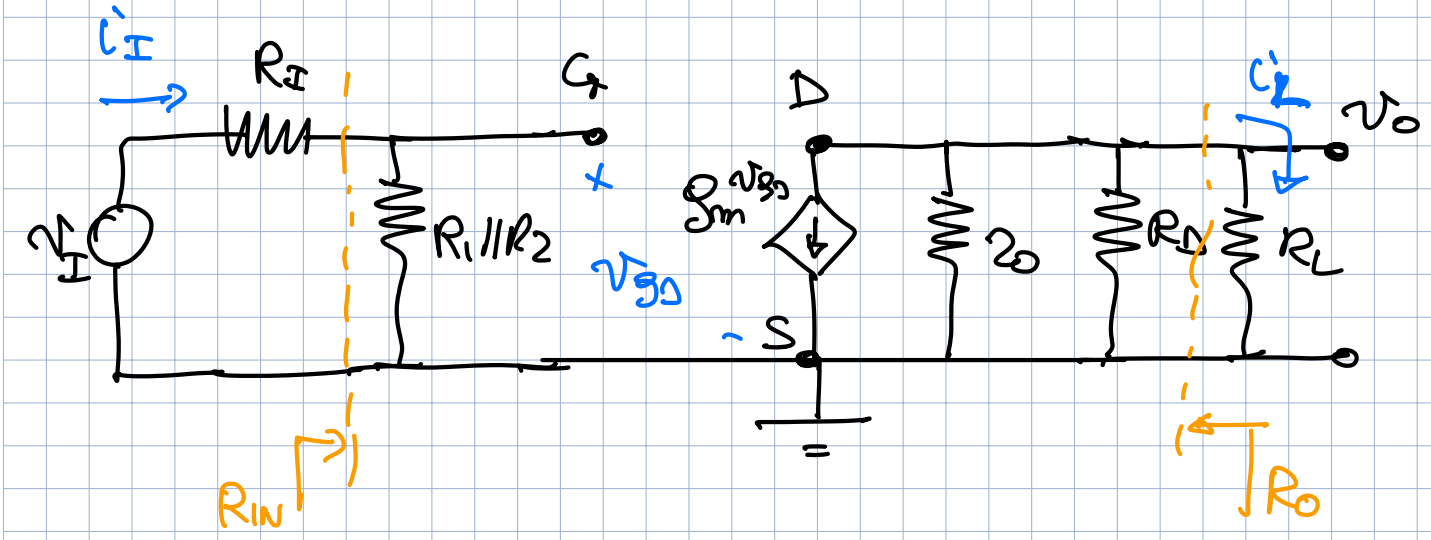
PARAMETRI PICCOLO SEGNALE:

$$g_m = \frac{2I_D}{V_{GS} - V_{TN}} = \frac{0{,}164\ mA}{0{,}81\ V} = 0{,}202\ mS$$

$$r_o = \frac{\frac{1}{\lambda} + V_{DS}}{I_{DQ}} = \frac{20 + 6{,}06\ V}{0{,}082\ mA} = 317{,}8\ k\Omega$$

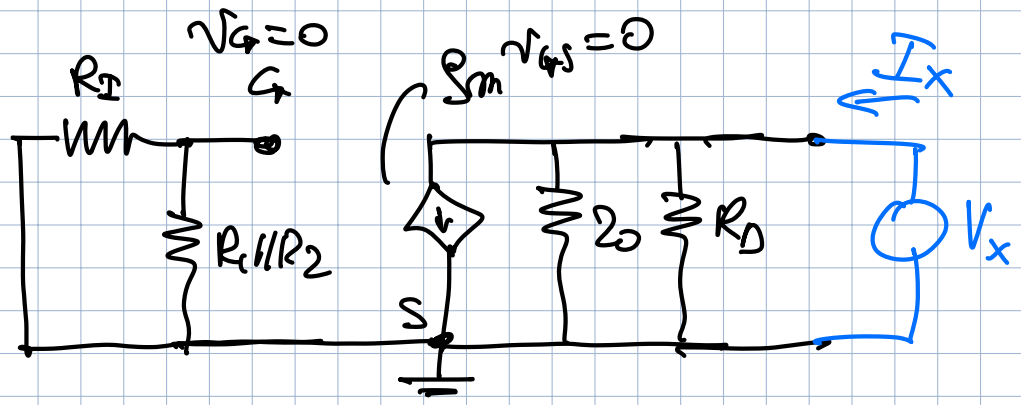
N.B. g_m è molto piccolo dovuto a correnti piccole $\Rightarrow$ serviva K_n + grande $\Rightarrow \frac{W}{L}$ più grande!

ANALISI AL PICCOLO SEGNALE



$$R_{IN} = R_1 // R_2 = 729{,}7 \ K\Omega$$

R_O ?



$v_G = 0$

$\Rightarrow g_m v_{GS} = 0 \quad \Rightarrow V_x = I_x (r_O // R_D)$

$$\Rightarrow R_O = \frac{V_x}{I_x} = r_O // R_D = 317{,}8 \ K\Omega // 82 \ K\Omega$$

$$R_O = 65{,}18 \ K\Omega$$

$$A_V = \frac{v_O}{v_I} = \frac{v_O}{v_G} \cdot \frac{v_G}{v_I} = -0{,}937$$

$$\frac{v_G}{v_I} = \frac{R_1 // R_2}{R_I + R_1 // R_2} = 0{,}999$$

N.B

$v_G = v_{GD}$

$$\frac{v_O}{v_G} = -g_m \underbrace{r_O // R_D // R_L}_{4{,}64 \ K\Omega} = 0{,}938$$

$$A_i = \frac{i_o}{i_I} = \frac{v_o}{R_L} \cdot \frac{1}{\frac{v_I}{R_I + R_1 // R_2}} = \frac{v_o}{v_i} \cdot \frac{R_I + R_1 // R_2}{R_L}$$

$$A_i = A_v \cdot \frac{R_I + R_1 // R_2}{R_L} = -136{,}94 \quad (*)$$

OPPURE CALCOLANDO PASSO PASSO:

$$i_L = -g_m v_{GS} \cdot \frac{r_o // R_D}{r_o // R_D + R_L}$$

$$v_{GS} = i_i \cdot R_1 // R_2$$

$$\Rightarrow A_i = \frac{i_L}{i_i} = -g_m R_1 // R_2 \cdot \frac{r_o // R_D}{r_o // R_D + R_L} = -136{,}94$$

SE RIPRENDO $(*)$

$$A_i = -\frac{R_1 // R_2}{\cancel{R_I + R_1 // R_2}} \cdot g_m r_o // R_D // R_L \cdot \frac{\cancel{R_I + R_1 // R_2}}{R_L}$$

$$= -g_m R_1 // R_2 \cdot \frac{r_o // R_D \cdot \cancel{R_L}}{R_o // R_D + R_L} \cdot \frac{1}{\cancel{R_L}}$$

$$= -g_m R_1 // R_2 \cdot \frac{r_o // R_D}{R_o // R_D + R_L} \qquad \text{C.V.D.}$$